Choupo

The Open Source Chemical Process Simulator

Explorer Guide

Choupo-dev

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1 What the Explorer is

The Explorer is Choupo's *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: **"what does this thermodynamics look like?"**

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway γ) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no "display model": if the Explorer shows an azeotrope at $x = 0.89$, the distillation column will hit it at $x = 0.89$.

2 The workflow

1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals, P^{sat}, c_p, formation — and what it lacks). Case-local components (from an open case's `constant/`) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state, P^{sat} correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

3 The views, one by one

3.1 Property vs T/P (scan)

Any pure-component property — P^{sat}, liquid density, c_p, viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

3.2 Pure phase diagram (P–T)

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

3.3 Binary boiling envelope (T–x–y)

The bubble and dew curves of a binary at fixed P. This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal → NRTL → UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

3.4 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary: γ_1, γ_2 against composition, with the infinite-dilution values γ^∞ at the edges. $\gamma > 1$: the pair dislikes mixing (positive deviation, minimum-boiling azeotropes); $\gamma < 1$: association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

3.5 McCabe–Thiele (distillation)

The classic x–y construction: equilibrium curve, operating lines, q-line, stages stepped off graphically. Change reflux and watch the stage count respond; push the specs toward an azeotrope and watch the pinch strangle the staircase. The fastest intuition-builder for distillation that exists — and it runs on the same K-values the rigorous column uses.

3.6 Binary flash (x–y + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy: $g_{\mathrm{mix}}(x)$ with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

3.8 Ternary boiling surface

T_{bubble} over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

3.9 Psychrometric chart

Humid air: humidity vs temperature with saturation, wet-bulb and enthalpy structure — the working chart for dryers and HVAC, computed from the same gas-phase thermodynamics as everything else.

3.10 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward SI $= 0$ (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

3.11 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the $T \times P$ plane. The whole point is the tension it makes visible: for ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first ("where do I write the reaction?" — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach ΔT (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts the $\Delta T = 0$ contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

3.12 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 — h_f, h_g, h_{fg}, entropies, volumes — the backbone of every steam and power-cycle case, browsable.

4 Provenance and honesty rules

The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

5 From Explorer to case: the hand-off

A typical session: you notice in the T–x–y view that your solvent pair azeotropes at 78 % — so the simple column in your flowsheet cannot deliver 99 % purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.